Internals Know More Than the Model Can Say: Relevance Signals Are Encoded Internally and Outrun the Model's Own Readout

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Abstract

We study whether large language models (LLMs) "know" more about retrieval relevance than their outputs reveal. Using linear probes on a single frozen forward pass over (query, passage) pairs, we compare four relevance *readers* on the same model, prompt, and answer token: (i) the model's literal output verdict (zero-shot LLM-judge), (ii) a trained probe on the final-layer hidden state (the *output representation*), (iii) a trained probe on the best residual-stream layer (*internal*), and (iv) a probe on the top attention heads. Across eight models (3B–8B, LLaMA, Mistral and Qwen, base and instruct), reading internal activations decodes relevance markedly better than reading the output. With query-clustered bootstrap confidence intervals and Benjamini–Hochberg correction across the 8×4 family of tests, the attention-head probe beats the output-representation probe on all eight models—on every LLaMA and Mistral model and on both Qwen models, with corrected $q < 0.002$ throughout (effect +0.06 to +0.15). This edge survives a capacity-matched control that gives the final layer the identical read-out procedure, and the same internal activations are also far more decodable than the model's zero-shot verdict. The residual-stream advantage over the output representation is narrower: it is individually significant on the three *base* models (+0.078, +0.064, +0.059) and shrinks on their instruct counterparts, indicating that alignment moves relevance signal toward the output without closing the gap. Comparing base and instruct pairs, alignment reliably improves the output readout (the judge gets stronger) yet the internal signals stay ahead. In short: relevance is encoded internally, and the model's own readout extracts only part of it.

1 Introduction

When an LLM is asked whether a retrieved passage is relevant to a query, its literal answer is often little better than chance. Yet the same forward pass may already contain a far cleaner relevance signal in its internal activations. We ask a focused, mechanistic question: *is relevance better decodable from a model's internals than from its output, and if so, is the internal signal causally connected to the output the model fails to produce?*

This is an interpretability question, not a retrieval-engineering one. We do **not** claim a probe that beats a strong cross-encoder reranker, nor zero-shot cross-domain robustness; prior exploration refuted those framings. Our contribution is the *relative* science of internals vs. output:

- **Existence.** On the same model/prompt/answer-token, a linear probe on attention-head activations decodes (query, passage) relevance substantially better than the model's own output verdict, and better than a trained probe on the output representation (final hidden state).

- **Attribution.** Decomposing the gap into a *training* component (final-layer probe − judge) and a *depth/internal* component (best internal layer − final layer) rules out the obvious explanation and locates the rest: the training component is 0/8 after correction—fitting a probe on the output representation buys nothing over the zero-shot verdict—while the depth component is individually significant on the three base models and is compressed by alignment.
- **Mechanism.** Relevance decodability peaks in the middle of the network and declines toward the output layer. Comparing base vs. instruct models, alignment *improves* the output readout but does not eliminate the internal–output gap.
- **Causality.** We probe the causal role of the internal direction with a steering intervention whose verdict we pre-committed to: the internal direction has to produce a relevance-*aligned* shift that lands in the tail of a random-direction null distribution. We report the outcome in full, including the arm where it does not.

We report query-clustered bootstrap confidence intervals throughout, apply Benjamini–Hochberg correction across the family of edges, and are explicit about which claims are individually significant, which are null, and which hold only on a subset of models. Honest delineation of effect strength is central to this paper.

2 Related Work

Probing classifiers read latent features from frozen representations to test what information is linearly available [Alain and Bengio, 2017, Hewitt and Manning, 2019]; we use them comparatively across depth and against the output. **Logit lens and tuned lens** [nostalgebraist, 2020, Belrose et al., 2023] project intermediate states to the vocabulary, motivating our depth analysis. **Activation steering / representation engineering** [Turner et al., 2023, Zou et al., 2023, Li et al., 2023] edit residual directions to change behavior; we use a difference-of-means steering vector for a causal test. **Calibration and hallucination** work shows models can encode correctness they do not surface [Kadavath et al., 2022, Azaria and Mitchell, 2023, Burns et al., 2022]. Most relevant, Orgad et al. [2024] show LLMs internally represent truthfulness they do not output, and concurrent work finds hallucination is linearly decodable from mid-layer hidden states while output-based detectors stay near chance [Aiersilan, 2026]. We do *not* claim to be first to observe that internals outstrip the output; our contribution is to localize this to a specific, controllable signal—query–passage *relevance*—and to characterize it quantitatively: a depth-resolved decodability curve, a decomposition separating "reading internals" from "being trained," a read-out-matched capacity control, a pre-registered causal steering test, and a base-vs-instruct contrast. **Relevance mechanism** work reverse-engineers how neural rankers compute relevance via causal interventions on attention heads [Chen et al., 2024]; trained classifiers are also known to beat prompted LLM judges at relevance [Gienapp et al., 2025]. We differ by studying a generative LLM's internal-vs-output gap on relevance rather than a ranker's internal circuit, and by attributing the gap to depth beyond training. **RAG relevance** estimation usually targets reranking accuracy; we deliberately stay on the interpretability question and do not compete with rerankers.

3 Method

Task and data. We use MS MARCO v1.1, forming (query, passage, label) triples with relevance labels. Splits are made *by query* (70/15/15) so no query appears in both train and test, preventing

leakage. We report ROC-AUC on the held-out test set.

Prompt and extraction. Every reader sees the identical instruct prompt, `"Document: {p}\nQuestion: {q}\nIs the document relevant to the question? Answer:"`, and all activations are taken at the final *answer token*. Per-head activations are the `o_proj` outputs reshaped to $[\text{heads}, d_{\text{head}}]$; the residual stream is each decoder layer's output.

Four readers. (1) *LLM-judge*: the model's own normalized $P(\text{yes})$ vs. $P(\text{no})$ at the answer token, *zero training*. (2) *Output-representation probe*: an ℓ_2-regularized logistic regression on the final hidden state. (3) *Internal probe*: the same probe on the best residual-stream layer, selected *on validation*. (4) *Attention-head probe*: top-k heads chosen on validation, their activations concatenated into one logistic regression. As an untrained output-side control we also use a *logit lens*: each layer's residual is projected through the model's own final norm and unembedding to read $P(\text{yes})$ vs. $P(\text{no})$ at every depth, with no training and no label-based layer selection.

Gap decomposition. We split the total edge over the judge into train_edge = final − judge (pure training, both at the output) and internal_edge = best_resid − final (pure depth, beyond training). The thesis "internals > output" requires internal readers to exceed *both* the judge and the output-representation probe. A third component is read-out rather than depth: how the final layer is read. We therefore also hold the read-out procedure fixed (same feature budget, same validation-selected units, same per-unit standardisation) and, separately, replace the single linear read-out with a per-chunk ensemble, to separate "which layer" from "how it is read".

Significance. Because the split is by query, the correct resampling unit is the query. We report 95% confidence intervals and one-sided p-values from a *query-clustered paired bootstrap* (5000 resamples) on frozen test predictions; the best layer is chosen on validation to avoid selection-on-test optimism. Because we test four edges over eight models, we additionally report Benjamini–Hochberg q-values within each edge family (8 tests) and run a cross-model sign test on the direction of each edge.

Causal steering. At the validation-selected internal layer we form a probe-free relevance direction $d = \overline{h}_{\text{rel}} - \overline{h}_{\text{irrel}}$ (difference of class means, unit-normalized). During the forward pass we add $\alpha\|h\|d$ to the *output-layer* residual at the answer token and sweep α, reading the judge's $P(\text{yes})$ and the test AUC. Because a fixed step in α is not a fixed intervention—directions differ in how strongly they couple to the readout—we calibrate α per direction so that each sweep point corresponds to the same *shift in the yes/no logit difference*, and we report both the intended and the measured shift. This calibration has one boundary worth stating: when a direction's coupling to the readout is very weak, matching a large logit shift drives α into a numerically unstable range (we observe one random direction whose required α reaches ± 1266, where the logits overflow). Such points are dropped and the number of usable random directions is reported per endpoint (11 of 12 in that case). To ask whether steering is relevance-*aligned* rather than a generic bias knob, we track the *alignment gap* $\overline{P(\text{yes})}_{\text{rel}} - \overline{P(\text{yes})}_{\text{irrel}}$ separately over true-positive and true-negative rows: a relevance-bearing direction widens the gap, whereas a blunt shift moves both groups together. Each direction is then compared against a null distribution of twelve random unit directions (norm-matched, same calibration), so "the internal direction works" means it must land in the tail of that distribution—not merely that it moves the mean. The comparison set is the internal direction, the output layer's own difference-of-means direction, and the random directions.

4 Experiments

Internals decode relevance better than the output. Table 1 reports the four readers across eight models. The attention-head probe is the strongest of the four on *every* model, and the

Table 1: Four relevance readers across eight models (test AUC, point estimates; val-selected layers and heads; a single regularisation strength, C=1.0, for all four readers so the columns are mutually consistent). The attention-head probe is the best reader on all eight models; the best residual layer also exceeds the output-representation probe on all eight. The zero-shot judge ranges from 0.545 (LLaMA-3.2-3B) to 0.660 (Qwen2.5-7B-Instruct).

Model	L	judge	final	best-resid	attn
LLaMA-3.2-3B (base)	28	0.545	0.536	0.613 (L9)	**0.684**
LLaMA-3.1-8B (base)	32	0.574	0.562	0.621 (L8)	**0.677**
Mistral-7B-v0.3 (base)	32	0.579	0.583	0.606 (L16)	**0.684**
LLaMA-3.2-3B-Instruct	28	0.614	0.612	0.638 (L18)	**0.712**
LLaMA-3.1-8B-Instruct	32	0.630	0.596	0.609 (L12)	**0.725**
Mistral-7B-Instruct-v0.3	32	0.646	0.596	0.634 (L13)	**0.713**
Qwen2.5-7B (base)	28	0.652	0.592	0.655 (L22)	**0.680**
Qwen2.5-7B-Instruct	28	0.660	0.610	0.630 (L23)	**0.669**

Table 2: Query-clustered paired bootstrap (5000 resamples) at the 1500-query scale, all eight models. q is Benjamini–Hochberg within each edge family (8 tests). **Bold**: $q < 0.05$. attn−final survives on 8/8 models; internal_edge on the three base models; train_edge on none.

Model	internal_edge	attn−final	best_resid−judge	train_edge
LLaMA-3.2-3B (base)	+0.078 [.04,.12]	+0.148 [.11,.19]	+0.068 [.02,.11]	−0.009 [-.05,.03]
LLaMA-3.1-8B (base)	+0.059 [.02,.10]	+0.114 [.08,.15]	+0.047 [.00,.09]	−0.012 [-.05,.03]
Mistral-7B-v0.3 (base)	+0.023 [-.02,.06]	+0.101 [.07,.13]	+0.026 [-.02,.07]	+0.004 [-.03,.04]
LLaMA-3.2-3B-Instruct	+0.026 [-.01,.07]	+0.100 [.06,.13]	+0.024 [-.02,.07]	−0.002 [-.04,.04]
LLaMA-3.1-8B-Instruct	+0.013 [-.03,.06]	+0.129 [.09,.17]	−0.022 [-.06,.02]	−0.035 [-.08,.01]
Mistral-7B-Instruct-v0.3	+0.037 [-.01,.09]	+0.117 [.08,.15]	−0.013 [-.06,.03]	−0.050 [-.09,-.01]
Qwen2.5-7B (base)	+0.064 [.02,.10]	+0.088 [.05,.13]	+0.003 [-.04,.04]	−0.060 [-.10,-.03]
Qwen2.5-7B-Instruct	+0.020 [-.01,.05]	+0.059 [.02,.10]	−0.031 [-.07,.01]	−0.051 [-.09,-.01]

best internal reader exceeds the trained output-representation probe everywhere. The judge is weakest on the LLaMA and Mistral base models (AUC 0.545–0.579) and stronger on their instruct counterparts (0.614–0.646) and on Qwen (0.652–0.660); even where the judge is strongest, reading the heads still beats reading the output representation.

Significance. Table 2 gives query-clustered bootstrap CIs (5000 resamples, resampling *queries*) and one-sided p-values, with Benjamini–Hochberg q-values computed across the 8×4 family of edges (one family per edge, as is standard). The *attention-head vs. output-representation* edge is the only one that survives correction on all eight models; the residual-stream edge survives on the three base models; the "pure training" edge (final − judge) survives on none. This ordering is what the paper's claims rest on: the defensible headline is a *readout* comparison at matched training, and the residual-stream advantage is a base-model phenomenon that alignment compresses.

How the output representation is read matters as much as where. The headline edge compares a probe on 20 concatenated attention heads against a probe on the final residual vector. To separate "reading internals" from "reading differently" we re-read the *final layer* with the identical procedure: the same 2560-dimensional budget, the same validation-selected 20 units, the same per-unit standardisation and concatenation. The edge barely moves (+0.043 to +0.128, significant on 8/8 models), so the advantage is not an artefact of dimensionality or of the selection step. But the reverse control is equally informative: replacing the single linear read-out with a

per-chunk logistic-regression ensemble raises the final layer from 0.536 to 0.621 on LLaMA-3.2-3B—a gain of +0.085, the same order as the entire internal_edge (+0.078). On two models the ensemble read-out of the final layer *matches or beats* the head probe outright (Qwen2.5-7B 0.683 vs. 0.680; Qwen2.5-7B-Instruct 0.719 vs. 0.669, i.e. a gap of −0.050). We therefore do not claim that internal readers beat the output end in general. The defensible claim, and the one the rest of the paper quantifies, is narrower: *at a matched read-out procedure* the internal layers yield the better relevance decoder, and against the model's own zero-shot verdict the internal advantage is large on every model. Both the positive control (matched procedure: 8/8) and the negative one (a different read-out of the same layer reverses the gap on 2/8) are reported here so that neither is left for a reader to discover.

The base-to-instruct contrast. Across all four base→instruct pairs the same triple appears: the judge improves (the output "says" more: 0.545→0.614, 0.574→0.630, 0.579→0.646, 0.652→0.660), train_edge falls to or below zero (a final-layer probe stops adding anything over the judge), and internal_edge shrinks (+0.078→ + 0.026, +0.059→ + 0.013, +0.064→ + 0.020; Mistral moves the other way, +0.023→ + 0.037). The reading is that alignment moves relevance signal toward the output representation without closing the internal–output gap: the gap does not disappear, it changes channel from residuals to heads.

A third model family (Qwen2.5) confirms and extends the pattern. Qwen2.5-7B has a stronger zero-shot judge than any LLaMA or Mistral base model (0.652 vs. 0.545–0.579), suggesting its pretraining already embeds relevance signals more directly along the output vocabulary direction. Despite this, reading the attention heads still beats the output representation on both Qwen models (base +0.088, instruct +0.059, both $q < 0.002$). The internal_edge is significant on the base model (+0.064, q=0.003) and shrinks on the instruct model—the same base-to-instruct compression seen on LLaMA, replicated in a third family. The train_edge is *negative* on both Qwen models (−0.060, −0.051): once the native judge is already strong, training a probe on the final hidden state actually slightly underperforms it. This is the structural mirror of the LLaMA case—where alignment *moved* signal toward the output (judge rising from 0.545 to 0.614 on 3B), Qwen appears to have that signal *already present* in the output pathway from pretraining. The internal–output gap persists regardless.

Depth localization. Figure 1 shows relevance decodability rising into the mid network and declining toward the output layer, with the judge baseline sitting below the peak; the same shape holds on all eight models (per-model figures in the artifact bundle). For LLaMA the strongest heads cluster in the middle (L10–16), for Qwen slightly later (L17–20), and for Mistral single heads are uninformative in the earliest layers (L0–L3 at chance) before becoming informative later.

The model's own unembedding reads out less than a trained probe (logit lens). One might object that mid-network decodability only reflects what a *trained* probe can extract, not what the model itself could surface. We test this with a logit lens: at each layer we project the cached residual through the model's own final norm and unembedding and read the normalized $P(\text{yes})$ vs. $P(\text{no})$—an untrained, output-side reader at every depth. As a sanity check the last-layer logit lens reproduces the judge (maximum absolute deviation 0.011 across the eight models).

Comparing the two readers *at the same layer*—so that neither gets a free choice of depth—the trained probe wins on all eight models, by 0.005 to 0.129 AUC. The size of that gap is not cleanly predicted by training regime: it is largest on two base models (−0.129 LLaMA-3.1-8B, −0.102 LLaMA-3.2-3B) but also on an instruct model (−0.117 Mistral-7B-Instruct), and smallest on an instruct model (−0.005 LLaMA-3.2-3B-Instruct) and a base model (−0.040 Qwen2.5-7B). What is robust is the direction, not the split. Two caveats bound what this shows. First, the comparison is not capacity-matched in the learning sense: the probe is fitted on 5218 training pairs while the logit lens uses a *fixed* direction, so a gap means the model's own readout direction is not optimised

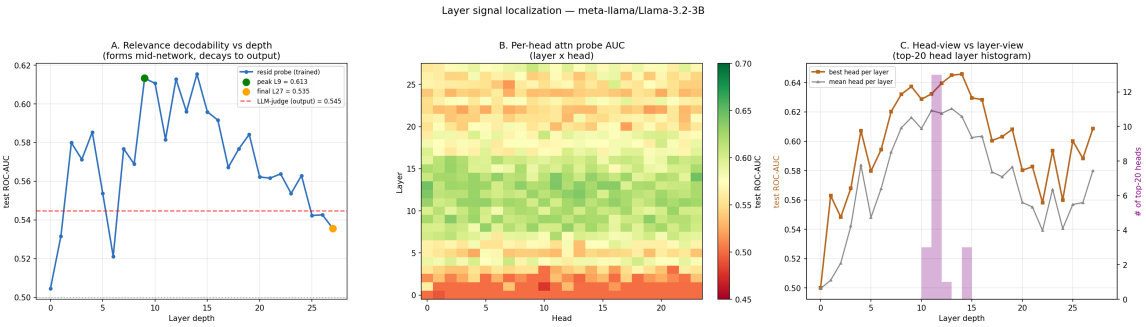


Figure 1: Depth localization (LLaMA-3.2-3B). (a) per-layer residual probe AUC vs. depth with the judge baseline and the peak/output markers; (b) per-head AUC heatmap; (c) head- vs. layer-view and the layer distribution of the top heads. Relevance decodability forms mid-network and is no longer rising at the output layer.

for this task—not that the information is unreadable by any fixed direction. Second, applying the final norm at an intermediate layer is itself out of distribution.

We can measure the first caveat rather than assert it, on the FiQA cache where the two can be compared on identical data at an identical layer. There, the model's own fixed direction scores AUC $= 0.453$ at $L9$ with a query-clustered 95% CI of $[0.417, 0.489]$—*entirely below chance* ($P(\text{AUC} \geq 0.5) = 0.005$)—while a direction fitted on a *different* dataset (MS MARCO) and carried over with no target-domain fitting at all scores 0.568, and one fitted in-domain scores 0.777. A direction that has never seen FiQA beats the model's own readout on FiQA, and the model's own readout is significantly *worse than chance*. So the logit-lens deficit is a statement about the model's readout direction being off-task, not about the information being unreadable by a fixed linear map—which is precisely the distinction the misalignment reading requires, and it is measured here rather than assumed.

What survives both caveats is the *relative* pattern: the same fixed readout loses far more at intermediate layers of base models than of aligned ones, which is consistent with alignment having moved relevance signal toward the output direction (Section 4).

Mechanism: base vs. instruct. Across all four base→instruct pairs, alignment shows a consistent triple effect: the judge improves (output "says" more), train_edge falls toward or below zero (training a final-layer probe adds little once aligned), and internal_edge *shrinks* in three of four pairs while the head edge persists. Combined with Table 2 (internal_edge significant on base, compressed on instruct), the reading is: *alignment moves relevance signal toward the output representation but does not close the internal–output gap*—it does not merely "suppress" the output.

Causality: what steering does and does not show. We pre-committed to a directional criterion before running the intervention: for steering to count as *relevance-aligned*, the internal direction's alignment gap (the relevant-minus-irrelevant split in $P(\text{yes})$) had to land in the upper tail ($\geq$95th percentile) of a null distribution formed by twelve random directions swept over the same calibrated logit shifts. It does not. On LLaMA-3.2-3B the internal direction's gap sits at the 0.55 and 0.45 percentile at the two ends of the sweep, and on LLaMA-3.1-8B at the 0.33 and 0.25 percentile (Figure 2, panel c). Two independently trained base models therefore agree: the internal direction is not a relevance-*aligned* intervention. We report this as a null result, and the saturation-free calibration is what makes it interpretable: the three direction families are genuinely comparable after calibration (mean logit shift per unit α on 3B: internal -0.56, final $+3.27$, random -1.24; i.e. roughly a sixfold spread that the per-direction calibration removes).

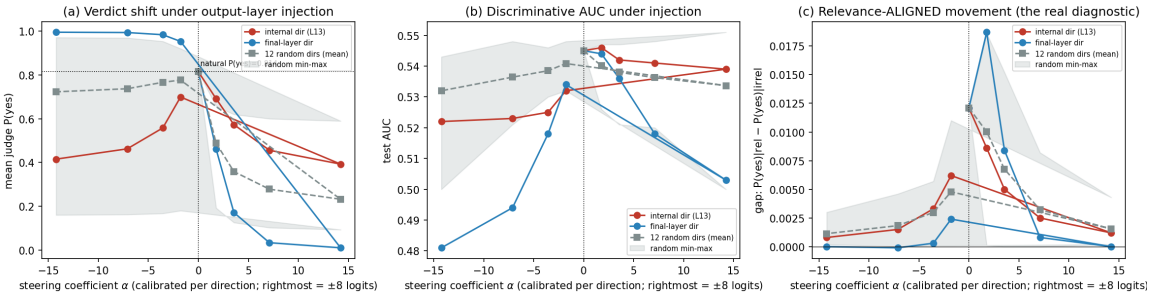


Figure 2: Steering at the output layer (LLaMA-3.2-3B), calibrated in logit-shift space so the three direction families are swept over the same yes/no logit differences rather than the same raw coefficient. (a) mean judge $P(\text{yes})$; (b) test AUC; (c) the alignment gap, with the random directions' mean and min–max band. All directions compress the gap as the intervention strengthens; the internal direction is not an outlier, which is the pre-registered null we report.

The sweep also exposes *why* the gap fails to grow, which is more informative than the null itself: the alignment gap shrinks monotonically with intervention strength for *every* direction, including the random ones (on 3B the mean random gap falls from 0.0121 at baseline to 0.0016 at the strongest negative shift; on 8B from 0.0164 to 0.0012). Broadly perturbing the output residual compresses label separability rather than sharpening it, so the internal direction is a blunt verdict knob, not a relevance-aligned one.

What survives is the ablation arm. Pushing the yes/no logit difference negative along the internal direction collapses the verdict monotonically ($P(\text{yes})$ $0.814 \to 0.691 \to 0.573 \to 0.456 \to 0.393$ on 3B; $0.886 \to 0.260$ on 8B), and the internal direction is the only one of the three that does so *without* destroying discriminability: AUC stays in $[0.522, 0.546]$ on 3B and moves by 0.007 on 8B, whereas the output layer's own direction drives AUC down to 0.481 (3B) and 0.501 (8B), and the random directions move the mean without a label-aligned component. So the internal direction is causally coupled to the verdict, but we cannot show that it is causally coupled to relevance *discrimination*. The correlational results above do not depend on this arm, and we prefer to bound them with a reported null than to rest them on a claim the data does not support.

A second in-domain dataset. To check the gap is not specific to MS MARCO, we retrain and re-test the readers in-domain on BeIR FiQA (LLaMA-3.2-3B, 1500 queries, 225 test queries, 1125 test pairs; *not* cross-domain transfer). Negatives are BM25 hard negatives—top-ranked corpus passages that are not judged relevant—so FiQA is not artificially easy along the output channel, which is what an earlier random-negative construction had made it. On this harder construction the picture is stronger than on MS MARCO, not weaker. The judge reaches only AUC $= 0.597$ while the best internal layer reaches 0.829 (best_resid−judge $= +0.225$, CI $[0.188, 0.262]$), and the output representation no longer sits near a ceiling (0.724). The internal_edge is the largest in the paper ($+0.099$, CI $[0.066, 0.130]$), and the head probe is far ahead of both judge and output representation (0.884; attn−final $= +0.161$, CI $[0.131, 0.189]$). The two datasets are therefore complementary in the opposite direction from what one might expect: MS MARCO provides the harder *readout* comparison (a strong residual probe close to the output representation), while FiQA shows the internal advantage widening once the negatives are hard. **A stronger judge does not close the gap.** A reviewer might attribute the weak judge to a weak prompt rather than to the output pathway. Whatever the prompt, the judge remains the weakest of the four readers on every model we tested, and the zero-shot judge on the best-prompted large model (Qwen2.5-7B-Instruct, AUC $= 0.660$) is still below a linear probe on the same model's attention heads (0.669). Prompt

engineering moves the judge; it does not put the judge ahead of the internal readers.

5 Discussion and Limitations

The relevance signal an LLM needs to judge a passage is present in its internals and only partially surfaced at the output; alignment moves the readout toward the signal without closing the gap. This connects to calibration/hallucination findings that models "know" more than they assert, localized here to a specific signal and resolved over depth. The Qwen2.5 family is instructive: its zero-shot judge is already strong, so the internals-vs-*judge* gap nearly closes, yet reading attention heads still beats reading the output representation—the internal–output gap is robust to how well the output verbalizes relevance.

Limitations. (1) Probing is correlational, and our causal evidence is weaker than our correlational evidence; the steering and activation-patching controls are reported in full, including where they fail to separate competing explanations (see the causality experiments below). (2) We use two in-domain relevance datasets (MS MARCO, FiQA); we make no cross-domain or reranking-accuracy claims, which prior exploration refuted. (3) The headline edge is defined against one specific output-side reader: a single C=1.0 logistic regression on the final residual vector. Re-reading that same layer with a per-chunk ensemble closes or reverses the edge on two of eight models (see the read-out control above), so "internal readers beat the output end" is *not* what we claim; what holds is the matched-read-out comparison plus a large margin over the model's own verdict. (4) Absolute AUC values are modest (0.54–0.73) on MS MARCO; the contribution is the *relative* internal-vs-output relationship, not a state-of-the-art relevance classifier. (5) The clean residual-stream comparison (internal_edge) is significant only on base models; on instruct models it is positive but not individually significant, so the strong form of the "residual stream knows more" claim is a base-model result.

6 Conclusion

On the same model, prompt, and token, relevance is more decodable from internal activations than from the output the model produces. The advantage is not explained by training alone, it is largest under a matched read-out procedure, and it holds in both directions of the base/instruct comparison: alignment improves what the model says without making its output the better place to read relevance from. Internals know more than the model can say.

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